



University of San Carlos - Talamban Campus

Department of Computer Information Sciences and Mathematics

In Partial Fulfillment of the Final Requirements for
IT 5101N - Business Analytics

IT 5101N Final Project - Phase 1

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Phase 1: Business Problem & Dataset Selection

Business Domain and Context

This project focuses on **e-commerce sales performance and customer purchasing behavior**. E-commerce businesses generate large amounts of transaction data from customer orders, product purchases, shipping activity, and regional sales. By analyzing this data, the business can better understand which products sell the most, which categories generate the highest revenue, how sales change over time, and which customer segments contribute to overall performance.

The purpose of this project is to use business analytics to turn e-commerce transaction data into useful insights. The analysis will focus on sales trends, product and category performance, regional patterns, customer demographics, shipping outcomes, and basic predictive relationships. These insights can help support better sales strategy, inventory planning, customer targeting, and operational decision-making.

Dataset Description

Item	Description
Dataset Name	E-Commerce Sales and Customer Insights Dataset
Source	GitHub: https://github.com/jeffersonmwatati/E-commerce-Sales-Analysis-Hackathon/blob/64527b0963ecab250ac30d546b2417a0cdfe6f42/raw-dataset-e-commerce-sales-and-customer-insights-dataset.csv
Format	CSV
Final Working File	Cleaned MySQL-ready CSV
Number of Records	1,000
Number of Attributes	20
Time Period	January 1, 2023 to January 3, 2024
Unit of Analysis	Each row represents one e-commerce transaction/order
Business Domain	E-commerce / Online Retail

The dataset contains transaction-level e-commerce sales records. It includes customer information, product details, order quantity, pricing, shipping fee, shipping status, and order date. The cleaned version also includes additional fields for age treatment and date-based analysis, such as `age_cleaned`, `age_group`, `age_model`, `year`, `month`, and `year_month`.

Data Preparation Summary

The data was cleaned using OpenRefine and additional validation checks to ensure that the dataset was suitable for business analytics and MySQL import. Before using the dataset for analysis, the file was cleaned and prepared to make it more suitable for business analytics and MySQL import. Column names were standardized into a database-friendly format, order dates were converted into a consistent YYYY-MM-DD format, and additional date fields such as `year`, `month`, and `year_month` were created to support trend analysis.

The Age column was treated carefully because some records had missing customer age values. Instead of deleting rows, the original age column was preserved, while a new `age_cleaned` column was created. Missing ages were recovered using the same Customer ID when possible. Remaining unresolved ages were placed into an Unknown age group for demographic analysis, while `age_model` was created for predictive modeling using median imputation. This allows the project to remain transparent while still supporting EDA, visualizations, and modeling.

Data Dictionary

Field Name	Data Type	Description
customer_id	Text/Categorical	Unique customer identifier. Customers may appear more than once if they placed multiple orders.
gender	Categorical	Gender of the customer.
region	Categorical	Region associated with the customer/order.
age	Numeric	Original customer age field. Some values are missing.
age_cleaned	Numeric	Cleaned age field after recovering missing values using the same Customer ID where possible.
age_group	Categorical	Age category used for customer demographic analysis.
age_missing_flag	Numeric/Flag	Indicates whether the original age value was missing.
age_model	Numeric	Age field prepared for predictive analysis using imputation where needed.
age_treatment_method	Text/Categorical	Explains how the age value was handled during cleaning.
product_name	Categorical	Name of the product purchased.
category	Categorical	Product category, such as Electronics, Accessories, or Wearables.
unit_price	Numeric	Price per unit of the product.
quantity	Numeric	Number of units purchased.
total_price	Numeric	Total transaction value.
shipping_fee	Numeric	Shipping cost charged for the order.
shipping_status	Categorical	Shipping outcome, such as Delivered, Returned, In Transit, or Unknown.

order_date	Date	Date when the order was placed.
year	Numeric	Year extracted from the order date.
month	Numeric	Month extracted from the order date.
year_month	Date/Text	Monthly period used for trend analysis.

Business Questions

Q1 — Descriptive Statistics

What are the key statistical properties of our numeric sales variables, and are there any notable anomalies or distributional patterns?

Before any serious analysis, the business needs a baseline understanding of its own numbers. How much does a typical order cost? Is pricing consistent or all over the place? Are there unusually large or small orders that could skew decisions? This question establishes the foundation everything else builds on.

Q2 — Trend Analysis

How do monthly total sales and order volume change across the available order period, and which full months show unusually high or low sales?

No business operates the same way every month. This question helps the business spot whether revenue is growing, shrinking, or seasonal — so they can plan inventory, staffing, and promotions ahead of time instead of reacting after the fact.

Q3 — Cross-Tabulation Analysis

How does total revenue vary across product categories and customer regions simultaneously?

Not every product sells equally well in every region. A category might dominate nationally but be weak in one region, which could mean a gap in marketing, logistics, or customer preference. This question finds those blind spots so the business can target efforts more precisely.

Q4 — Drill-Down Analysis

Starting from total sales, how does performance break down by region, category, and product-level performance?

Overall sales figures can show whether the business is performing well, but they do not explain where revenue is actually coming from. A strong sales month may be driven by only a few regions, categories, or products, while other segments contribute very little. By drilling down from overall sales into region, category, and product-level performance, the business can identify which areas consistently drive revenue and which segments may require operational, inventory, or marketing improvements. This analysis helps decision-makers understand whether growth is broad across the business or concentrated in specific products and markets.

Q5 — Correlation and Predictive Analysis

Which numeric variables most predict total order value, and can a simple model quantify that relationship?

Understanding what drives revenue lets the business make smarter decisions. Should they push higher-priced products or focus on selling more units? The model gives a concrete, data-backed answer — and lets them estimate expected revenue for any given order scenario.

Ad Hoc Q6 — Follow-Up from Q2

For the lowest full sales month, which products, categories, or regions contributed most to the weaker monthly performance?

Spotting a dip in Q2 is useful, but knowing what caused it is what makes it actionable. If Electronics dropped, that's a supply or demand issue. If Accessories dropped, it might be a promotion gap. Without this follow-up, the business knows something went wrong but not what to fix.

Ad Hoc Q7 — Follow-Up from Q3

For the strongest region-category sales combination, is performance mainly driven by higher order count, higher average transaction value, or specific products?

A strong region-category combination may perform well because customers place more orders, purchase higher-value products, or respond better to specific product offerings. If Electronics dominate in a particular region, the success may not simply come from demand volume alone — it may be driven by premium products such as laptops, smartphones, or monitors generating larger transaction values. This question helps the business determine whether strong performance is primarily caused by order frequency, high-value transactions, or specific products, allowing the company to make better decisions on inventory planning, product focus, regional strategy, and sales targeting.

Scope of the Analysis

This project focuses on descriptive business analytics using cleaned e-commerce transaction data. The analysis will examine sales performance, product/category behavior, regional differences, customer demographics, shipping outcomes, and basic predictive relationships. The goal is to provide insights that can support sales strategy, inventory decisions, customer targeting, and operational improvements.